



ALGORITHM DESIGN

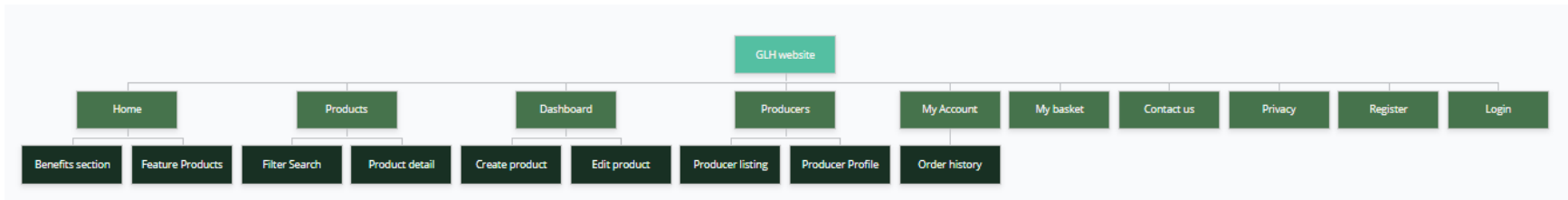
Nelson Ogoto

GREENFIELD LOCAL HUB

Contents

Site Map	2
.....	2
Styles	2
Dark Mode	2
Background	2
Font	3
Interactive	3
Buttons/Cards	3
Light Mode	3
Background	3
Font	3
Interactive	3
Accent Colours	3
Visual Designs	3
Home Page	3
Products	5
Producer Dashboard	5
My Account	6
Contact Us page	6
Privacy Policy page	7
Basket page	8
Producers page	8
Algorithm Design.....	8
Log In	8
Register	10
Filter Products	11
Add a product to basket.....	12
Edit Availability as Producer.....	13
Database	14
Entity Relationship Diagram.....	14
Data Dictionaries.....	14
ASPNETUSERS.....	14
ASPNETROLES.....	15
ASPNETUSERROLES	15
ASPNETROLECLAIMS	15
ASPNETUSERCLAIMS.....	15
ASPNETUSERLOGINS	15
ASPNETUSERTOKENS	16
PRODUCTS.....	16
ORDERS	16
ORDERPRODUCTS	17
DISCOUNTS.....	17
BASKET	17
BASKETPRODUCTS	17
PRODUCERS.....	18
Test Strategy	18

Site Map



- Home Page is the main landing page that every visitor would first encounter. It contains the hero section, The “why buy local” benefits, featured products and a preview of the producers.
- Products is the full product catalogue that showcases all the products sold buy various producers with a search bar and filter menu
- Producers is a page dedicated to showing all the producers that are selling their products on the website
- Dashboard is a page for users with the producer role. They are able to see all the products of the producer and pending orders containing the products of the producer
- Create product is a page for users with the producer role to add a new product
- Edit product is a page for users with a producer role to edit the details of a product
- My account is a page where users can manage their account as well as view their order history
- My basket contains all the products that the user has added to their basket with a checkout button to order all the products in the basket
- Contact us is a page for users to get in touch with the company customer service
- Privacy Is the legal compliance page. It outlines how user’s data is being used and protected
- Register page allows users to register an account in the company
- Log in page allows users to log in to the website

Styles

Font Family = “Inter”

Dark Mode

Background

- Body: #1F2A24 
- Foreground Elements: #2F3E35 

Font

- Normal: #F5F5F5 
- Lighter: #C8D5C0 

Interactive

- Normal: #6BBF59 
- Lighter: #4F9F4A 

Buttons/Cards

- Corner radius: 30px (±10px)
- Shadow: 10px blur drop shadow
- No outline

Light Mode

Background

- Body: #FAF7F2 
- Foreground Elements: #FFFFFF 

Font

- Normal: #2B2B2B 
- Lighter: #6B6B6B 

Interactive

- Normal: #2F6F4E 
- Hover/Selected: #1F4F36 

Accent Colours

- Loyalty & discounts: #F4A259 
- Success: #4CAF50 
- Warnings: #E9B44C 
- Errors: #D9534F 

Visual Designs

Home Page

This is the top announcement bar. This displays free delivery promo and keeps users aware of key offers. Colour scheme is dark green and the font is 13px sans serif regular white.

This highlights the brand name and home link. The icon was chosen to give an organic feel to the website. The colour used is green for the icon and the font is 14px for the bold name and 9px for the tag

Nav links for site navigation and way finding. The colour used is #1a4a1a active underline, font is 16px regular and 500 on active. Active state is bottom borderline

The search bar is used to search up products quickly, the font is 14px regular placeholder and it uses a modern, friendly Rounded pill

The Basket button is used to access the cart at all times and show live updates when an item is added. Colour is #639922 pill along with white counter. Font is 16px bold white text

The Accessibility bar is used to showcase all the WCAG support tools. Font is 14px bold white text

The Primary CTA "Shop products" is used to drive users to buy products sold. colour is #1a4a bg, white text. Font is 16 px weight and 500 pill. It has strong contrast

The secondary CTA "Meet Producers" is used to direct all users seeking info about the producers to the producers page. Colour is transparent bg with dark border. Font is 16px regular pill

The Hero Illustration is used to reinforce the farm identity of the company. The font is 12px caption below the image. Colour is #c8e6a0 bg, #63922 plant SVG.

The Stats such as the "12+ producers" "80+ products" "2hr collection" and "5mi" vg distance are all used to enhance the websites appeal of the website to the user by showing relevant information that would reassure the user. Font is 20px bold value and 14px label

The Hero headline colour is #1a4a1a darket green. It highlights the company name and its font is 18px bold

The hero subtext is used to expand briefly on the hero headline. The font is 14px regular line-height. Colour is var(-color-text-secondary)

The section header is used to show featured products that would entice the user to purchase, colour used #1a4a1a, 18px weight 500

The product image zone is used to show a visual of the product identity. The colour used is #eaf3de light green bg. Font is 28px emoji placeholder

The view all link is used to rout users to the products page. Colours used are #639922 green. Font is 14px regular plus arrow. Its subtle and non comping with the header

In Stock badge is used to identify if a product is in stock and available at a glance. The colour used is #EAF3DE light green bg, #386D11 text. Font is 11px bold pill badge

Product name is used to identify the product as its the primary product label. The colour is var(-colour-text-primary). Font size is 14px weight 500

Price is used to show the amount for each item. The colour used is #1a4a1a dark green, Font is 14px weight 500

The add to basket button is the button primarily used for purchasing items. The colour used is #639922 bg, white text. Font is 14px weight 500 pill.

The loyalty banner will be used to promote loyalty signap as the high contrast block grabs attention. Colour is #1a4a1a dark green bg and Font is 16px bold heading, 12px body.

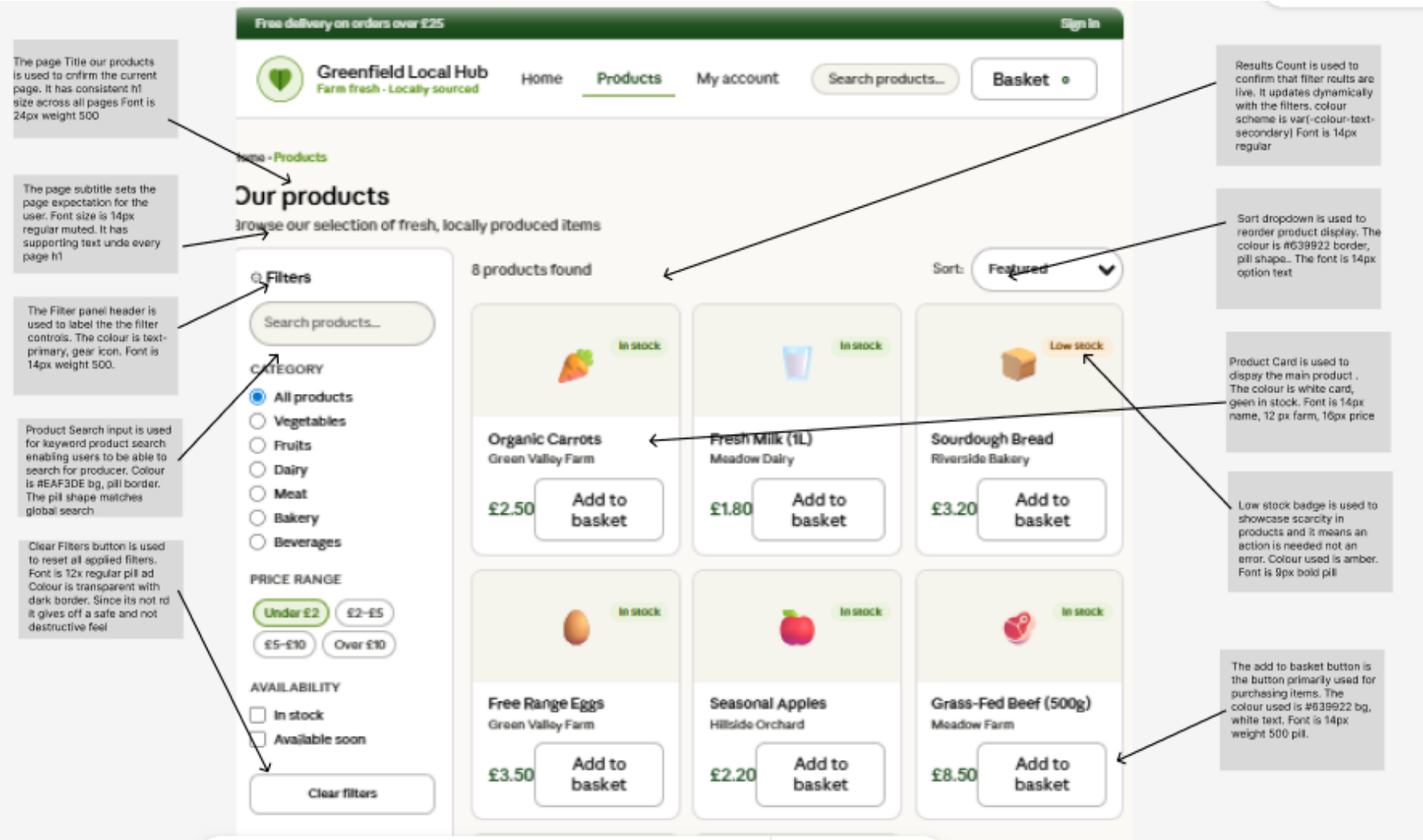
The loyalty points counter s used to personalise the banner value for the user. A large number creates excitement. Colour used is #a8dg78 light green number. Font 22px bold, 11 px muted label

The Producer Cards are used to showcase the producer's identity. Colours used are #c8e6a0, #d4eecc lighter green, #ffe8b0 warm amber. Font is 16px name bold, 12px type muted.

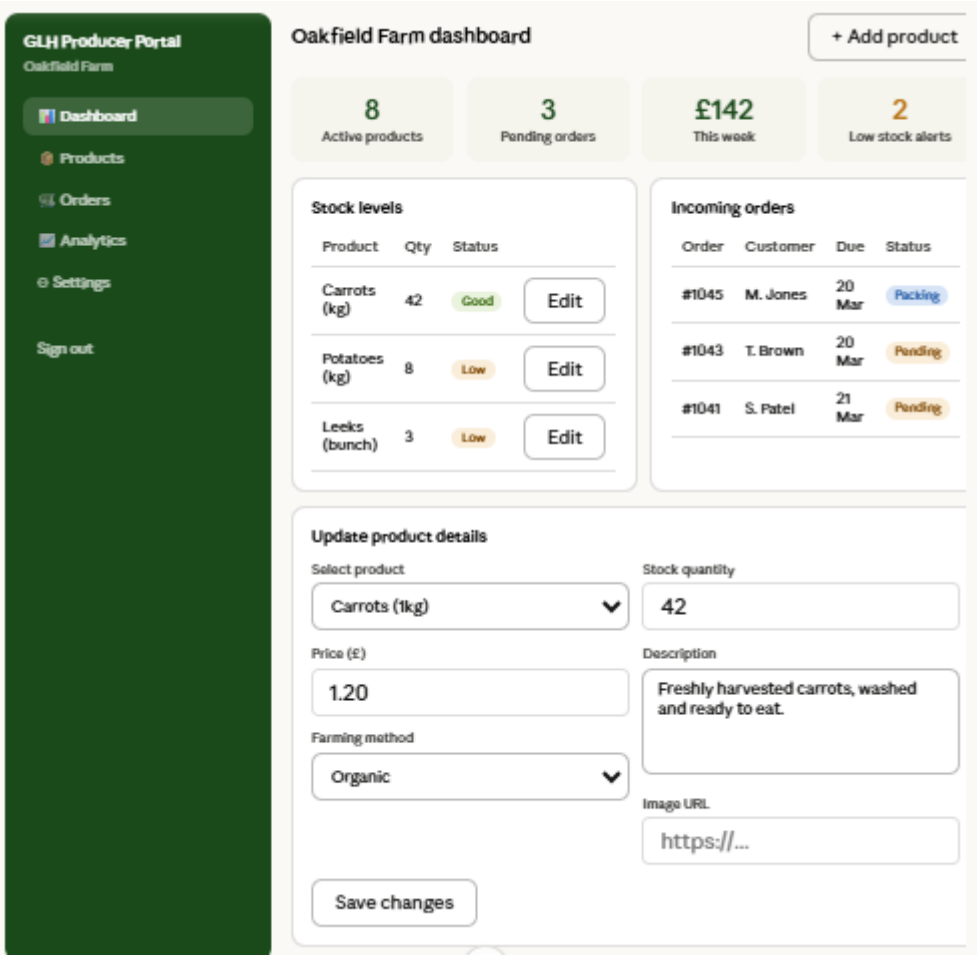
The View Products Button is used to enter the products page. The colour used is transparent bg, dark border, Font used is 14px regular full width pill.



Products

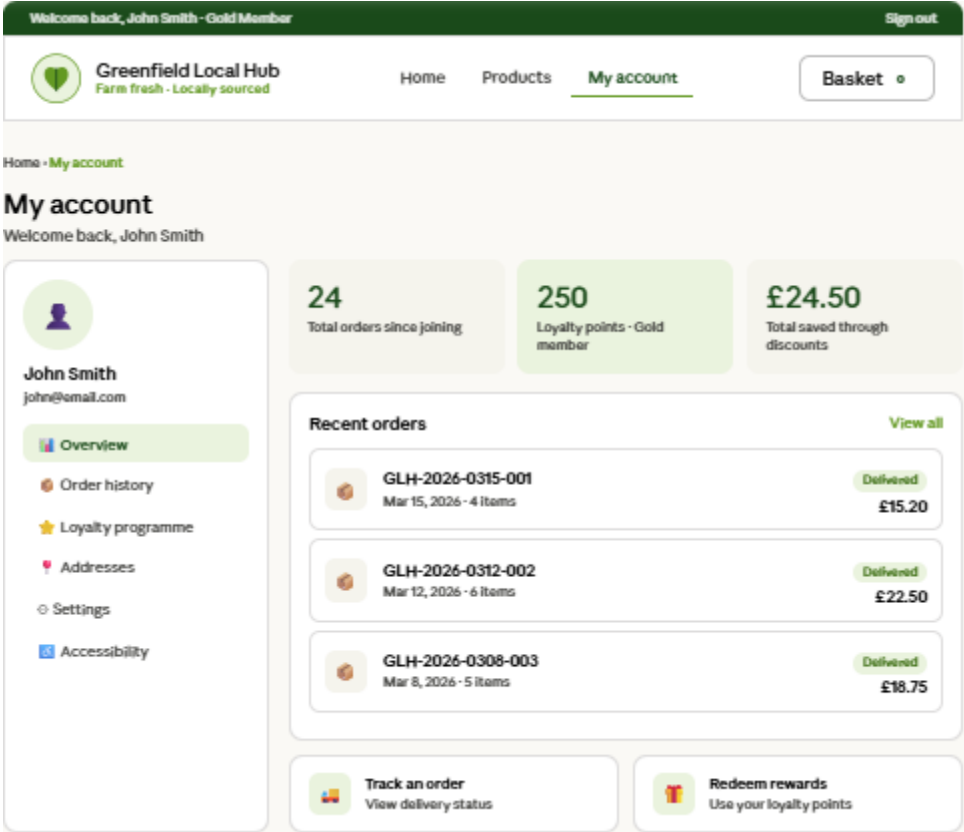


Producer Dashboard



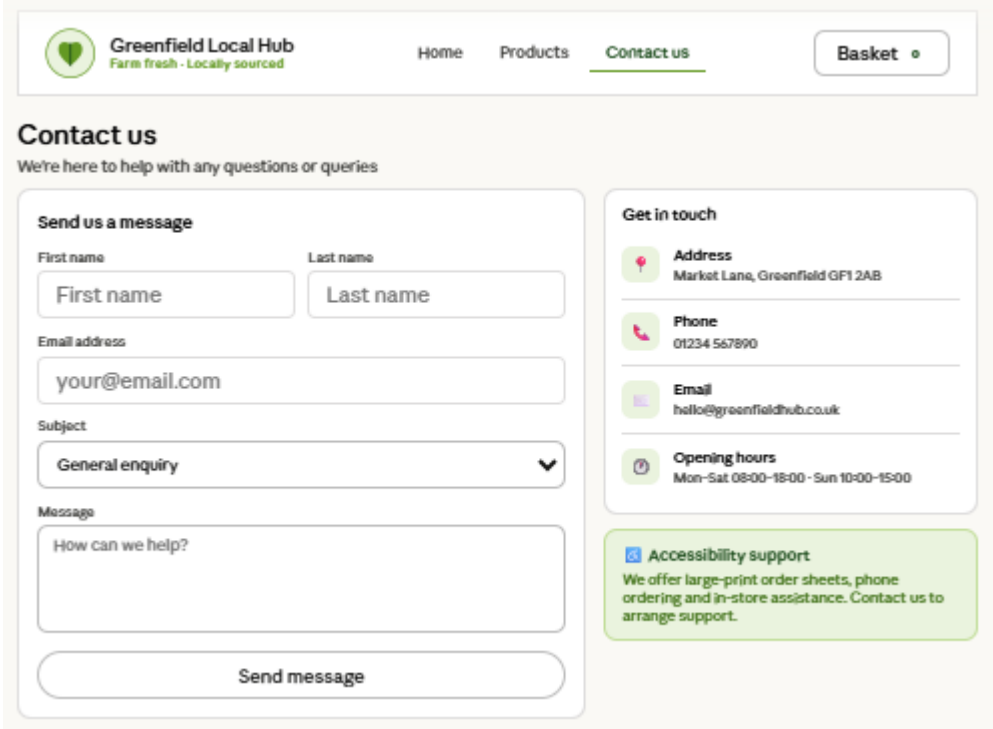
- The dark side bar GLH portal is used to back-office brand Identity. The dark tone highlights that it's a producer portal and not a shop. The colours used is #1a4a1a bg with a white text. Font is 14px bold name, 12px sub muted
- The sidebar Nav(Dashboard (active)) is used for the navigation of the producer section. Font is 12px regular, icon prefix. The colour is rgba(255, 255, 255,.15). Once its active it has a subtle white bg fill.
- Sidebar Nav- (Analytics/settings). This is used for the data and configuration sections. It has a consistent icon-prefix navigation pattern. Font is 12px regular. The colour is rgba(255, 255, 255,.7) muted
- Sidebar Nav(Products/Orders) is used for other portal section links. The colour is rgba(255, 255, 255,.7) muted. Font is 12px regular
- Sign Out link (sidebar bottom) is used to log out from producer portal. Font is 12px and the colour is muted white. Its placed at the bottom to avoid accidental clicks from users.
- The Add product button is used to create new product listing. It's the primary dashboard action located on the top right. Font is 12px weight 500 pill. It has a #639922 bg, white text
- The Stock levels table is used to manage product quantities. Font is 12 px table, 10px badg4e, 10px button. The colour is white card, row borders 0.5px

My Account



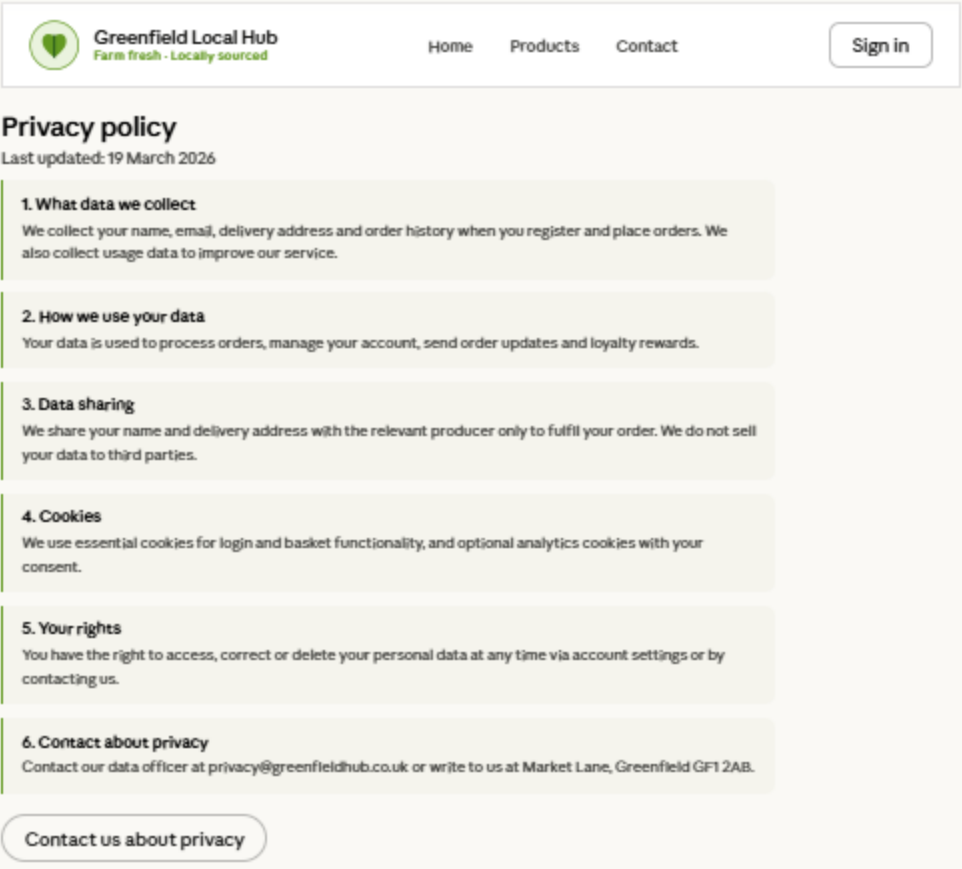
- The username(John Smith) is used to confirm the logged in identity of the user. The name and email is stacked below the avatar circle. Font is 14px weight 500
- Side Bar Nav(Loyalty Programme)- This is the points and rewards section. The star emphasizes the rewards concept. Font is 12px regular
- The stat card(Total Orders:24)- This shows the total number of orders since joining . A large number gives a sense of achievement. The Font is 22px bold value, 12px label. The colour is #EAF3DEBG, #1A4A1A Value.
- The stat card(Loyalty Points:250)- This shows the current points balance. The colour is #EAF3DEBG, green value. The Font is 22px bold, 12px label below
- The stat card(Total saved: £24.50)- This shows the real money benefit and reinforces the loyalty scheme. The Font is 22px bold, 10px label. The colour scheme is #EAF3DE bg, #1a4a1a value.
- The Order Row- This shows individual order summary. The colour scheme is a white row, green badge. Fonts is 14px id bold, 12 px meta muted.
- Delivered status badge- This shows the order status at a glance. Green means it was completed positive state. The font is 9px bold green pill. The colour is # EAF3DE bg, #3B6D11 text.

Contact Us page



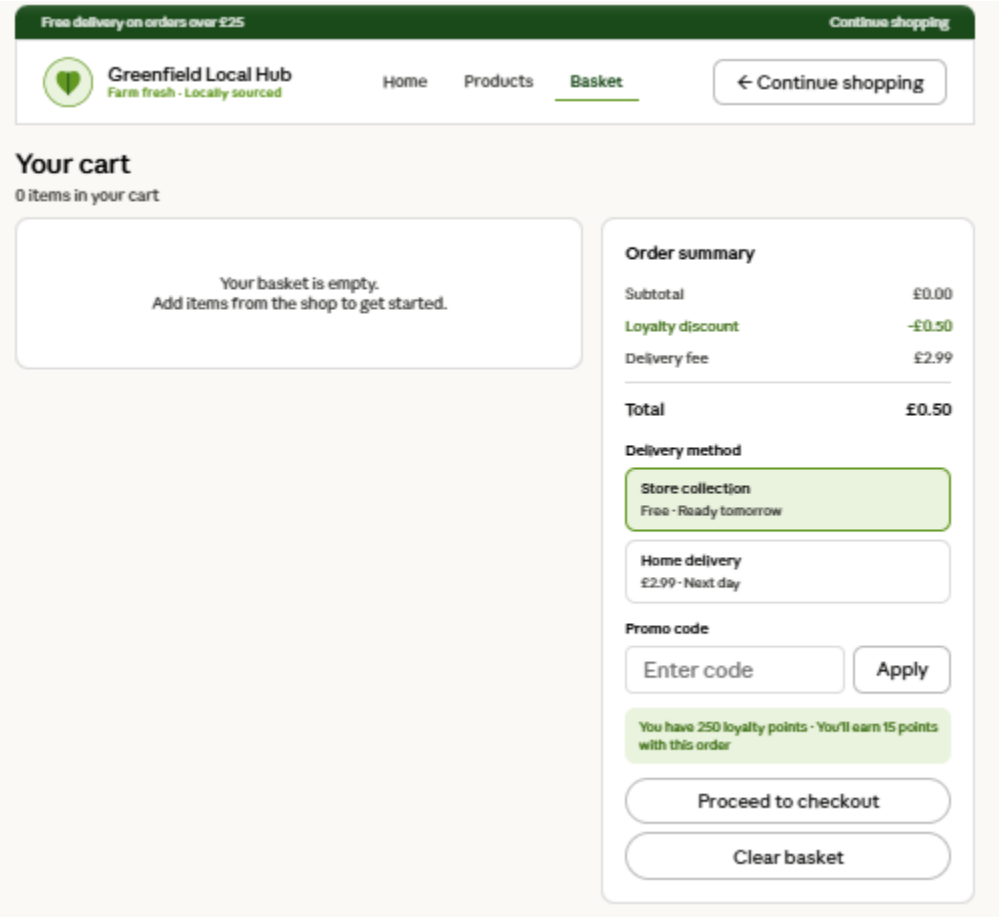
- First and Last name inputs- This is used to identify the enquirer. The font is 14px input, 12px label muted. The colour scheme is white input, border-secondary. It's a split grid between first name and last name
- Email Address input- This is used to capture reply address. The font is 14px input, 12px label. The colour scheme is white input.
- Send Message button- This is used to submit the contact form. Font is 13px bold pill, 100% wide.
- Get in touch Panel header. This labels right info column. Font is 12px bold. The colour is text primary.
- Address, Phone, Email Info Item- This highlights the different ways of getting services from the company. Font is 12 px label bold, 10px value. Colour wscheme is #EAF3DE for the icon bg, #3B6D11 icon and this is consistent for all of them.

Privacy Policy page



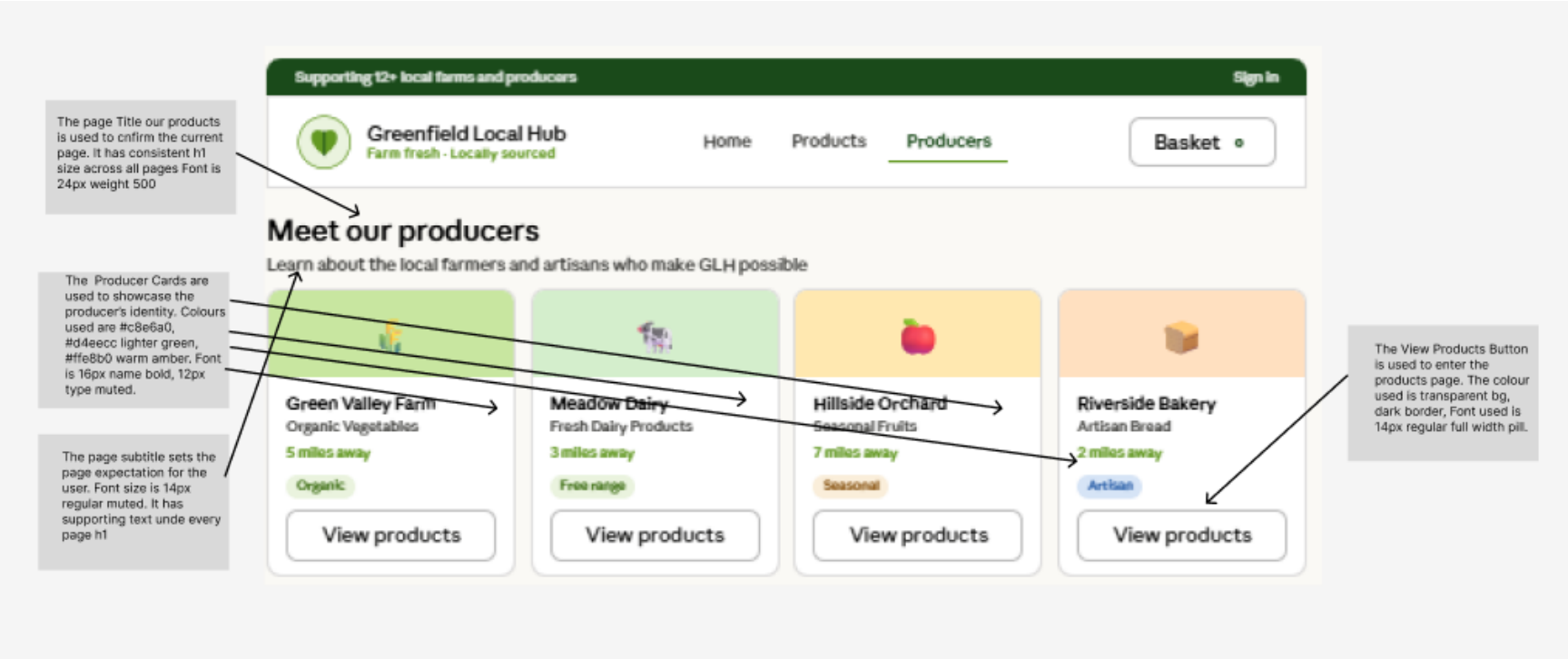
- Logo + Nav(Privacy page)- The purpose of this is a consistent navigation bar. The colour is #639922 icon and dark text. The font is 14px bold logo, 12px nav links.
- Sign in button (Top right)- This is used to log in on or log on legal pages, the font is 12px regular. The colour scheme is a transparent background with a dark border. No basket shown means there’s no shop content.
- Page Title(Privacy Policy)- This confirms the legal page context. The font is 20px weight 500. The colour is the primary colour text.
- Last Updated Date- This is a GDPR-required date stamp. The Font is 12px regular muted and the colour is the primary colour text.
- Sections 1-6- These are all required information set in the website to inform the user of the privacy policies the company adheres to.

Basket page



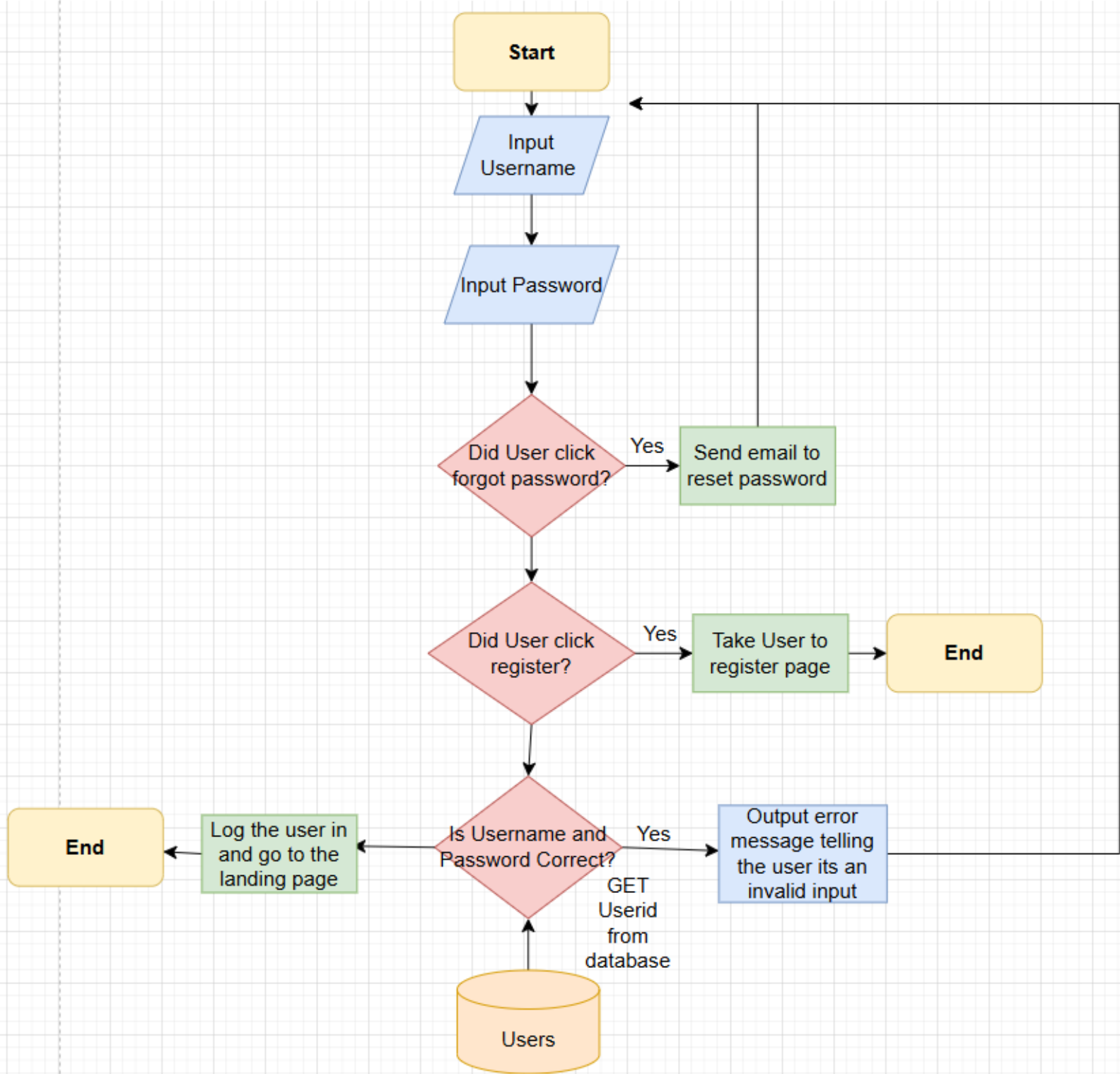
- Page Title- This confirms the user is in the basket. The font is 20px weight 50. The colour is primary text colour
- Item Count(0 items in your cart)- This shows the live basket summary line. It updates when items are added and removed. The font is 12px regular muted. The colour is primary text colour.
- Basket item image- This is used to visually confirm the correct item. In the real build it will have an actual product photo. The font is 22px emoji stand in. The colour is #EAF3DE bg placeholder.
- Item Name- This identifies the product. The font is 12px weight 500. The colour is the primary text colour,
- Clear basket button- This removes all items at once. The font is 12px, it matches the checkout width. The colour is a danger red pill button.
- Delivery Method selector- This highlights the two methods of delivery (collection or delivery). The font is 11px title bold 10px subtle and the colour is #EAF3DE selected highlight.
- Order summary Panel- This shows a checkout overview. The right column always stays visible. The font is a 13px heading, 11px row text. The colours are a white card, 0.5px border.

Producers page



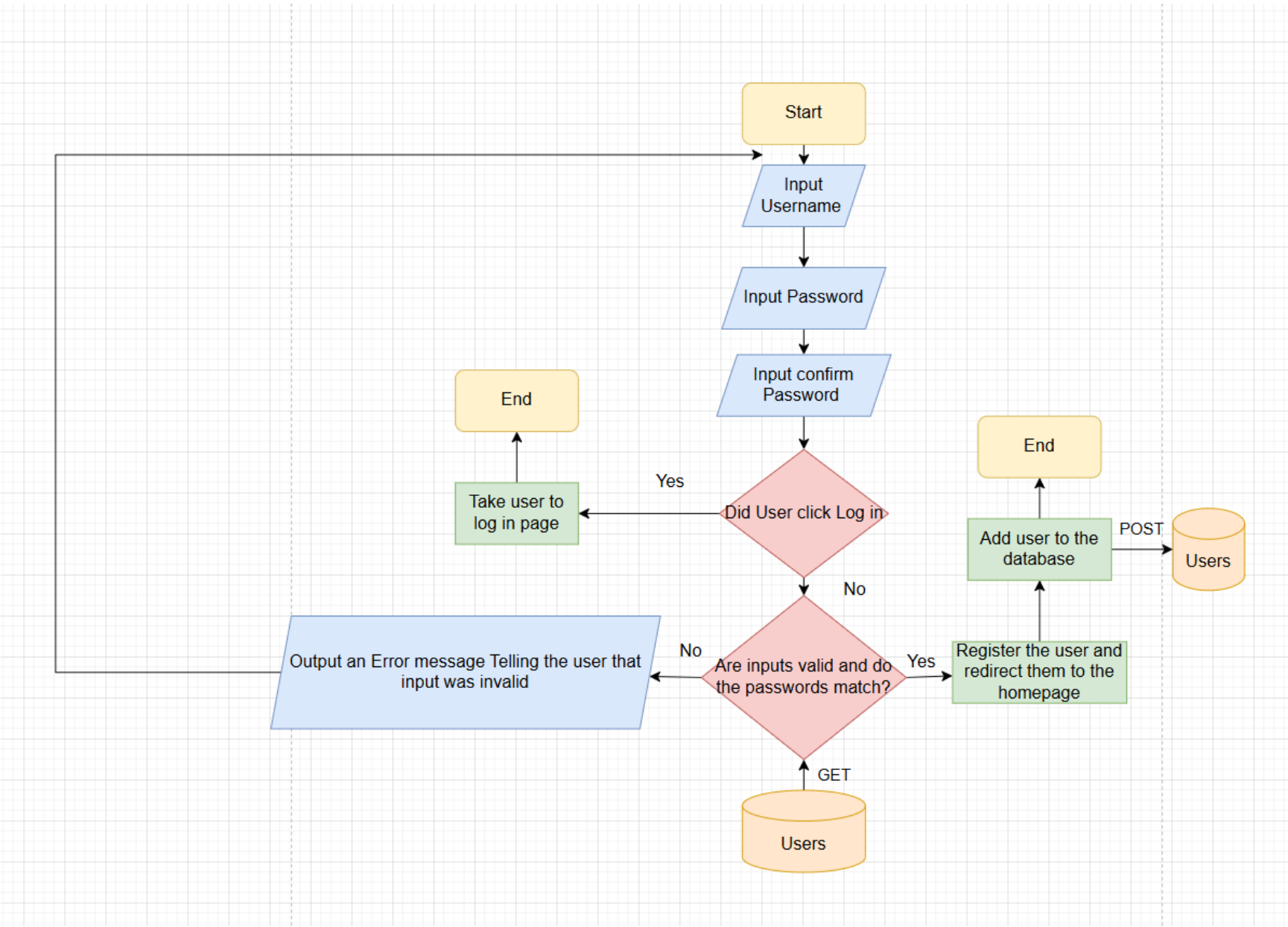
Algorithm Design

Log In



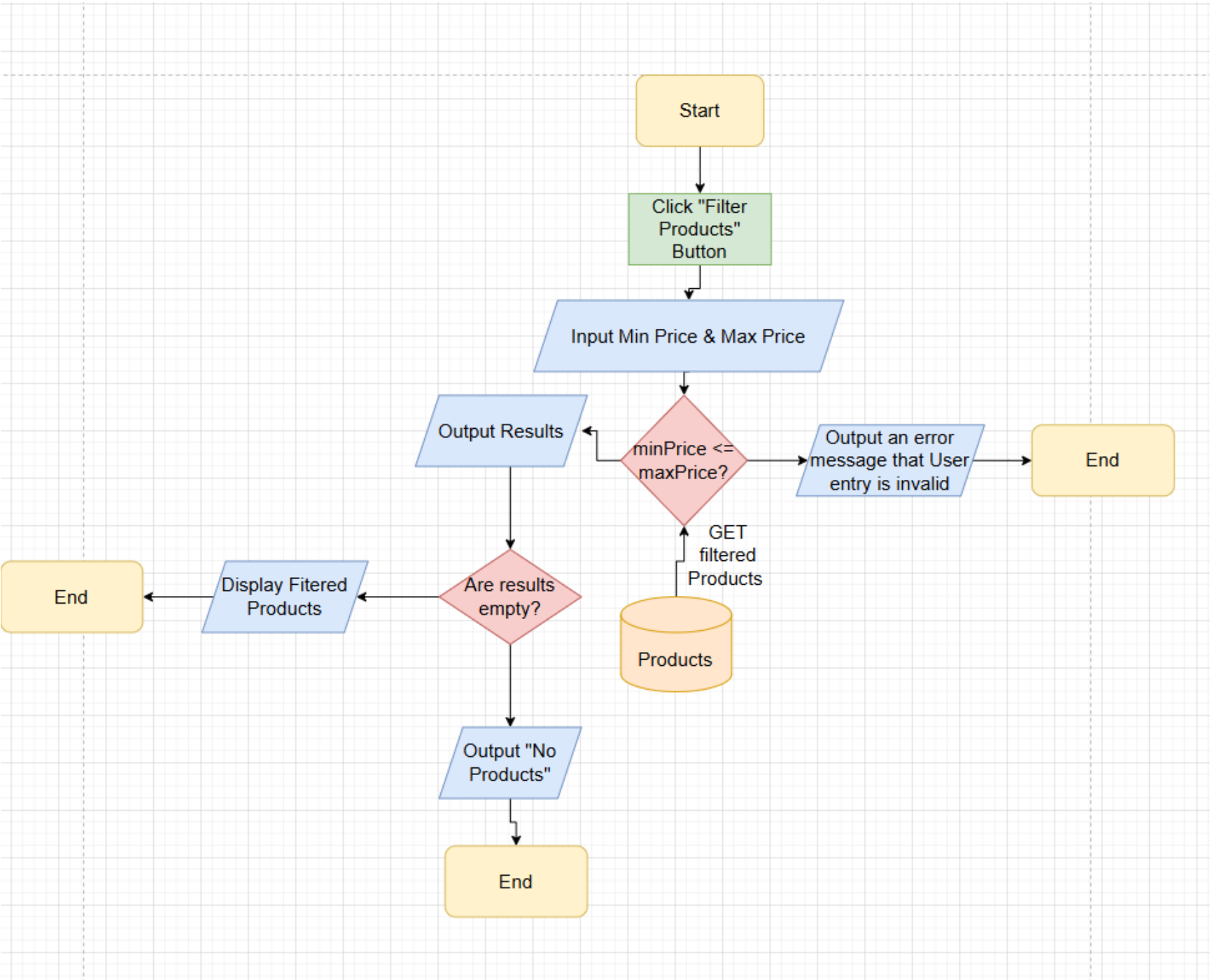
The user will be prompted to input their username and passwords then the system will check if the user has made actions like clicking forgotten password, if this is the case the system will send a password reset email to the user. If the user clicks the register button the system will redirect them to the register page. If the user doesn't click any of those the system will check if the password and username is valid, if its not the system will output an error message that the user log in info is in invalid thus sending them back to re-enter their log in info. If the user information is correct the system will log the user in and redirect them to the home page.

Register



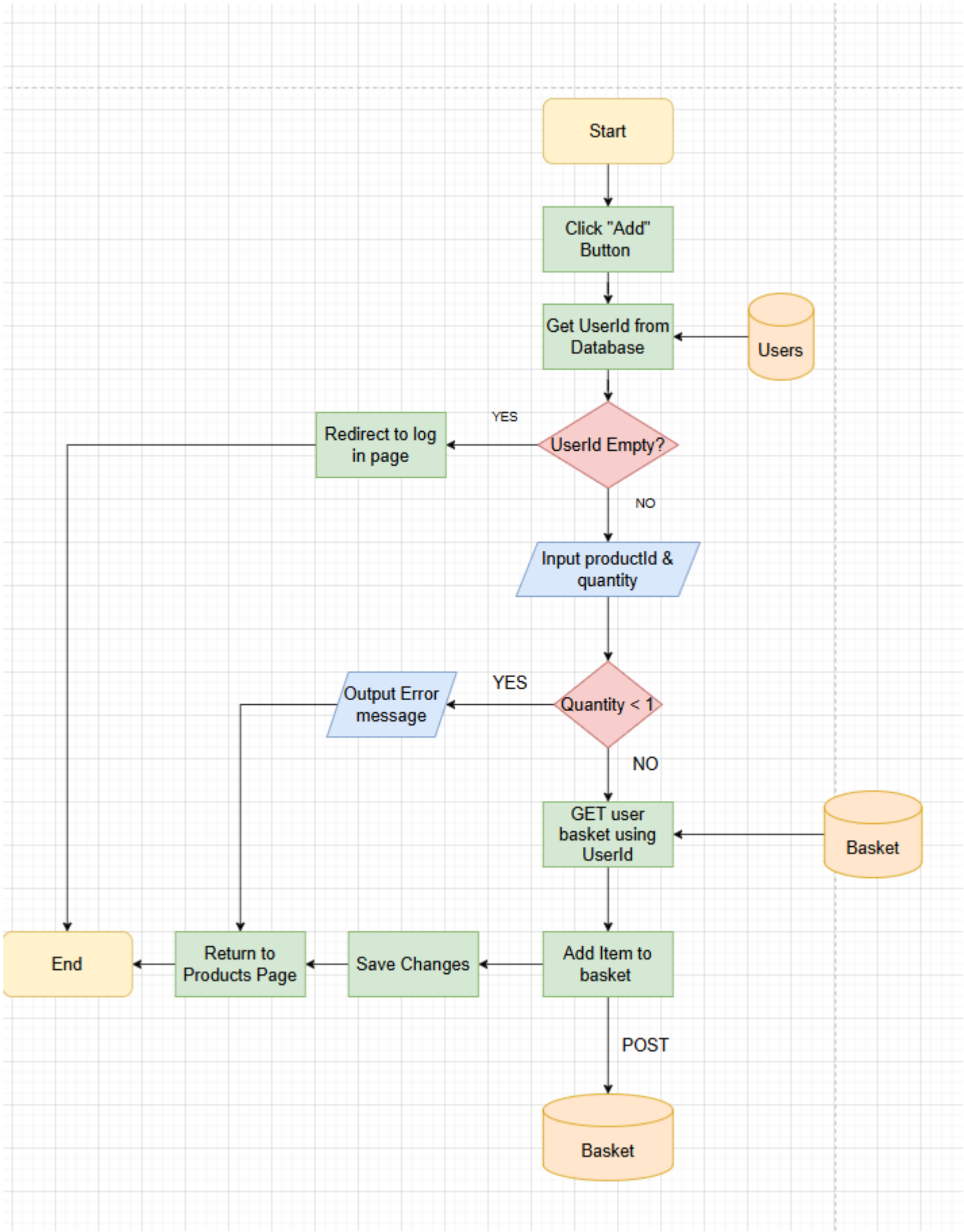
The user will be prompted to input their username and password and also to confirm the password. If the User interacts with the log in button, the system will redirect the user to log in page. If not then the system will check if the user inputs were valid. If the inputs were not valid the system outputs an error message informing the user that the input is invalid and prompts them to re-enter their log in details. If the inputs were valid then the system registers the user and adds them to the database.

Filter Products



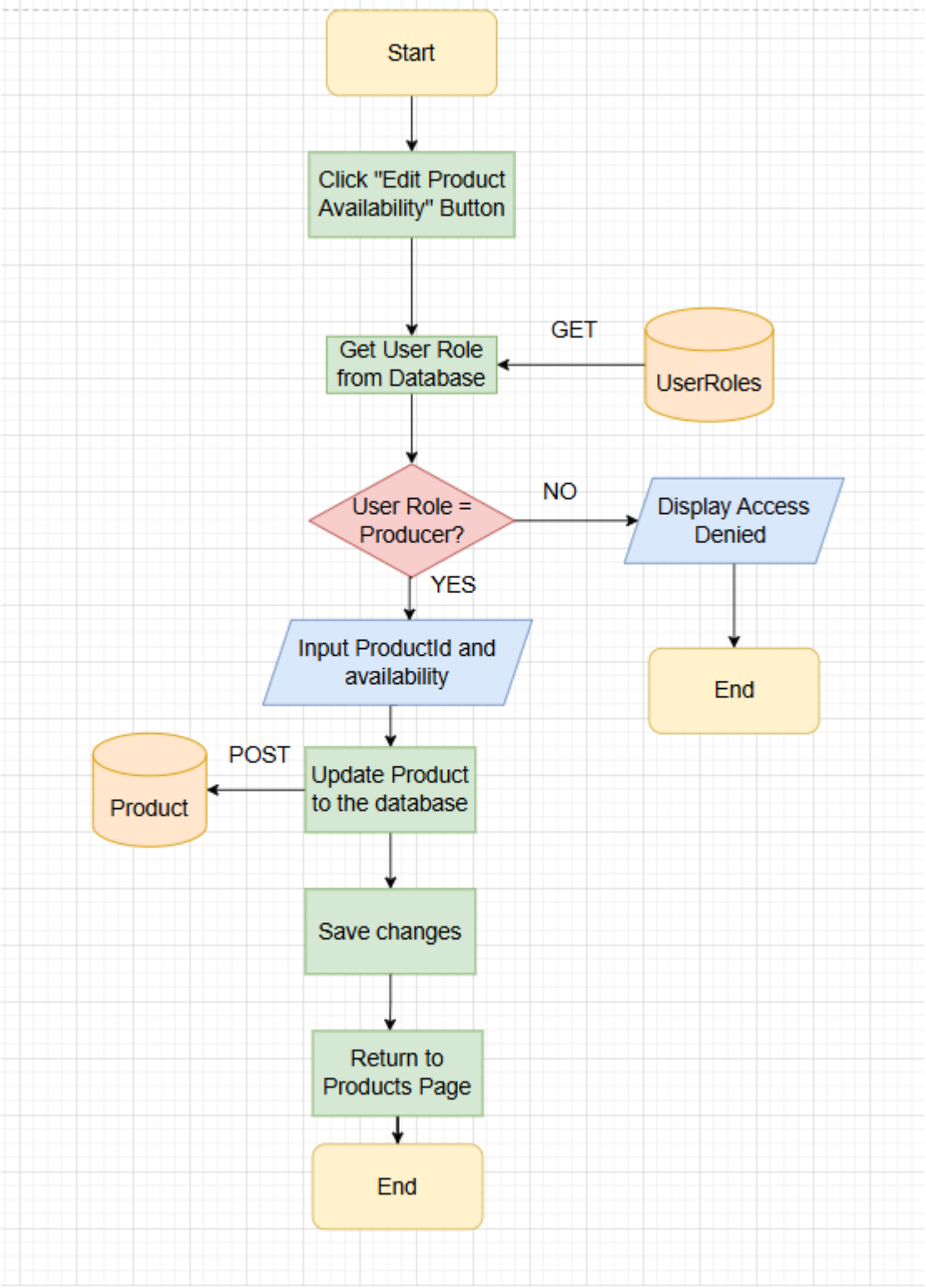
The user will be prompted to click the filter products button and also input minimum and maximum price. If the minimum price is not less than or equal to max price then the system outputs an error message of invalid entry. If the minimum price entered is less than or equal to max price then the system will output the results. If the results are empty then the system outputs “No Products” and the program ends. If the results are not empty then the system will display the Filtered products.

Add a product to basket



The User clicks the “Add” button then system gets the userId from the database. If the userid is empty then the user is redirected to the log in page. The user is then prompted to input the ProductId and the quantity of product. If the quantity is less than 1 the system outputs an error message and the user returns to the Products page. If the quantity is not less than 1, the system gets the user basket using the userId from the database and Adds the item to the basket. The system then saves the item in the database then returns to the product page.

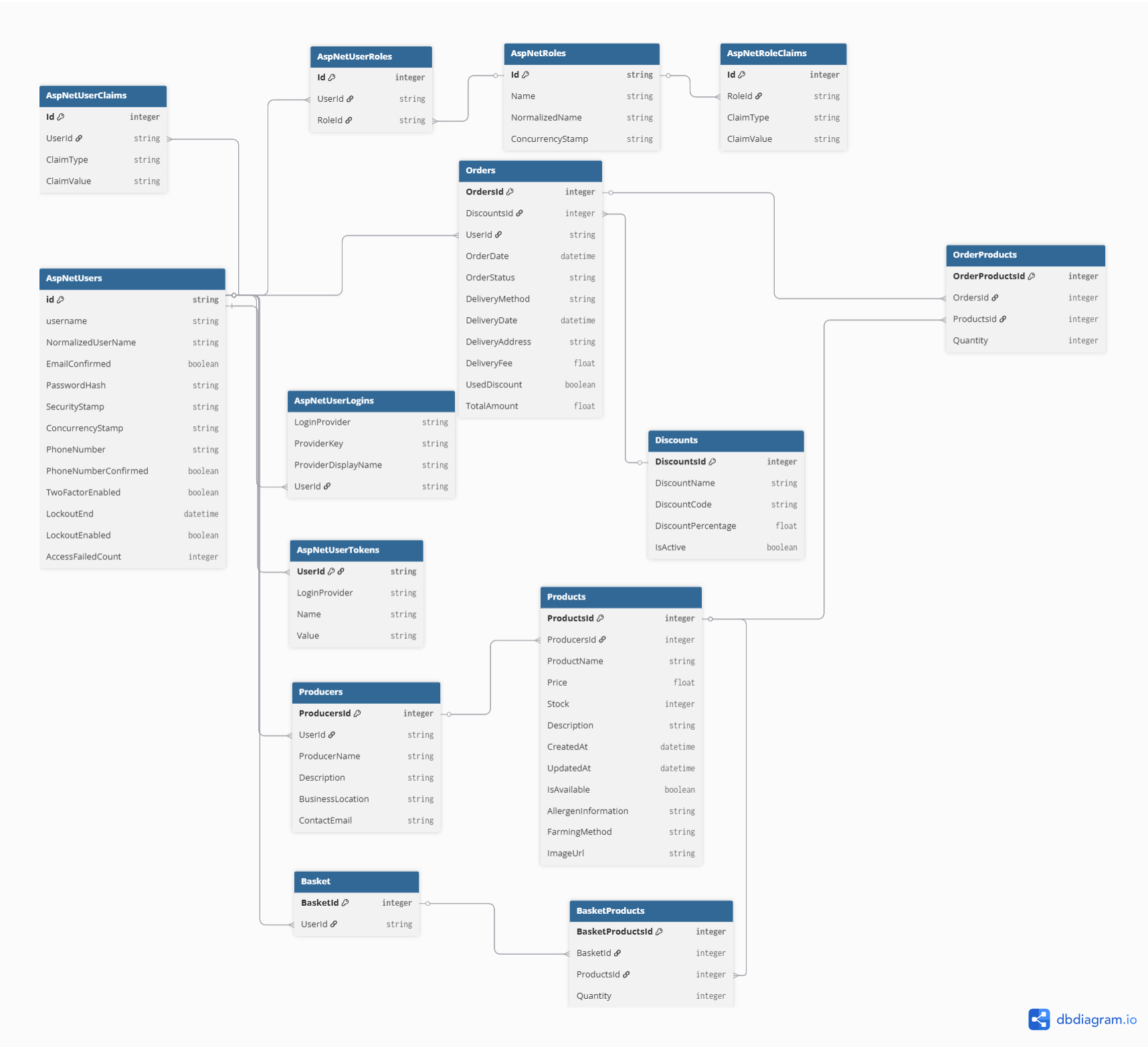
Edit Availability as Producer



When the user clicks the “Edit Product Availability button” the system retrieves the User role of the user. If the user is not a producer then the system outputs a message of denied access. If the user role is a producer then the user is prompted to input productid and availability. The product is then updated to database (POST). The system saves the changes then returns the user to the products page.

Database

Entity Relationship Diagram



Data Dictionaries

ASPNETUSERS

Column	DataType	Required	Default	Validation	Key Type
Id	String	Yes			Primary
UserName	string	yes			None
NormalisedUserName	String	Yes			None
Email	string	Yes		Must include a valid email structure, must have “@” symbol, must have	None

				a domain name.	
Normalised Email	String	Yes			None
EmailConfirmed	Bool	Yes		Users must confirm their email	None
PasswordHash	String	Yes			None
SecurityStamp	String				None
ConcurrencyStamp	String				None
PhoneNumber	String				None
PhoneNumberConfirmedTwoFactorEnabled	Bool				None
LockoutEnd	Bool				None
LockOutEnabled	Datetime				None
AccessFailedCount	Int				None

ASPNETROLES

Column	Data type	Required	Default	Validation	Key Type
Id	string	Yes			Primary
Name	String	Yes	Users Id		None
NormalizedName	String	Yes			None
ConcurrencyStamp	String	Yes			None

ASPNETUSERROLES

Column	Data type	Required	Default	Validation	Key Type
Id	int	Yes			Primary
UserId	String	Yes	Users Id		Foreign
RoleId	String	Yes	RolesId		Foreign

ASPNETROLECLAIMS

Column	Data type	Required	Default	Validation	Key Type
Id	int	Yes			Primary
RoleId	String	Yes	RoleId		Foreign
ClaimType	String	No			None
ClaimValue	String	No			None

ASPNETUSERCLAIMS

Column	Data type	Required	Default	Validation	Key Type
Id	int	Yes			Primary
RoleId	String	Yes	RoleId		Foreign
ClaimType	String	No	No		None
ClaimVaue	String	No			None

ASPNETUSERLOGINS

Column	Data Type	Required	Default	Validation	Key Type
LoginProvider	String	Yes			Primary
ProviderKey	String	Yes			None
ProviderDisplayName	String	No			None
UserId	String	Yes	Users Id		Foreign

ASPNETUSERTOKENS

Column	Data Type	Required	Default	Validation	Key Type
UserId	String	Yes	Users Id		Foreign
LoginProvider	String	Yes			None
Name	String	Yes			None
Value	String	No			None

PRODUCTS

Column	Data type	Required	Default	Validation	Key Type
ProductsId	int	Yes			Primary
ProducersId	int	Yes			Foreign
ProductName	String	Yes			None
Price	Float	Yes		Must be greater than 0	None
Stock	Int	Yes		Must be greater than 0	None
Description	String	Yes			None
CreatedAt	datetime	Yes			None
UpdatedAt	Datetime	Yes			None
IsAvailable	Boolean	Yes			None
AllegenInformation	string	Yes			None
FarmingMethod	string	Yes			None
ImageUrl	string	Yes		Url must link to a vaid image	None

ORDERS

Column	Data type	Required	Default	Validation	Key Type
OrdersId	int	Yes			Primary
DiscountId	int	No	DiscountId		Foreign
UserId	String	Yes	UserId		Foreign
OrderDate	Datetime	Yes		Date entered must not be today's date. Order date must be filled in	None
OrderStatus	String	Yes	Pending	Must only be a set order status option	None
DeliveryMethod	String	Yes		Must only be Pickup or Delivery	None
DeliveryDate	Datetime	Yes			None
DeliveryAddress	String	Yes		Must only be available if	None

				Delivery on the method of picking up orders was chosen. Must be a valid address.	
DeliveryFee	Datetime	No		Must only be available if Delivery on the method of picking up orders was chosen	None
UsedDiscount	Boolean	Yes	False		None
TotalAmount	Float	Yes		Must not be less than or equal to 0	None

ORDERPRODUCTS

Column	Data type	Required	Default	Validation	Key Type
OrderProductsId	int	Yes			Primary
OrdersId	String	Yes	OrdersId		Foreign
ProductsId	Float	Yes	ProductsId		Foreign
Quantity	Int	Yes		Must be greater than 0 and must not be greater than what's in stock	None

DISCOUNTS

Column	Data type	Required	Default	Validation	Key Type
DiscountsId	int	Yes			Primary
DiscountName	String	Yes			None
DiscountPercentage	Float	Yes			None
IsActive	Boolean	Yes			None

BASKET

Column	Data type	Required	Default	Validation	Key Type
BasketId	int	Yes			Primary
UserId	String	Yes	Users Id		Foreign

BASKETPRODUCTS

Column	Data type	Required	Default	Validation	Key Type
BasketProductsId	int	Yes			Primary
BasketId	Int	Yes	BasketId		Foreign
ProductsId	Int	Yes	ProductsId		Foreign
Quantitiy	Int	Yes		Must be greater than 0	None

PRODUCERS

Column	Data type	Required	Default	Validation	Key Type
ProducersId	int	Yes			Primary
UserId	String	Yes	Users Id		Foreign
ProducerName	String	Yes			None
Description	String	Yes			None
BusinessLocation	String	Yes		Can only be cities in England	None
ContactEmail	string	Yes			None

Test Strategy

Date of test	Component to be tested	Type of test to be carried out	Prerequisites and dependencies
20/4/26	Log in	Whitebox testing, Black box testing, Functionality, Security Integration	The log in page must be created and the migration and update-database must have been completed. The controller logic must have been added to handle user errors. Test Data: Normal Email: manual@gamil.com Password: Password123! Erroneous Email: noemail Password: 678 Extreme Email: manual@gamil.com Password: musta3# (Exactly 6 characters which is the minimum)
20/4/26	Register	Whitebox testing, Black box testing, Functionality, Security Integration	Register page must be fully functional. The migration and update-database must have been completed. The controller logic must have been added to handle user errors. Test Data: Normal Email: manual@gamil.com Password: Password123! Erroneous Email: noemail Password: 678 Confirm Password: 678 Extreme Email: manual@gamil.com Password: musta3# Confirm Password: musta3#
20/4/26	Log Out Function	Black box testing, Functionality testing, Security Integration	No test data needed. The Log in system should be fully functional and allow

			users to create accounts and save their data to a database and the log out button must be visible to the user.
20/4/26	Link buttons on the navigation bar and the footer	Black box Functionality Integration	All the pages for each link must be completed and each link must be functional. The Navigation bar must be made and fully functional. Tester must check each link to see if its functional and leads to the correct page
20/4/26	Accessibility features	Black box Functionality Integration	Accessibility features must be fully functional and accessible to the user. Tester should check each accessibility feature to test if its functional. No test Data needed.
20/4/26	Cookies Functionality	Black box Functionality integration	Cookies must be made. Tester should check if the cookies show up when new users enter the website. No test data required
20/4/26	Home Page	White box, Black box, Functionality Integration	Home Page must be fully functional. Tester must be able to access other pages from the home page No test data needed.
20/4/26	Placing an Order	Blackbox Testing, White box Testing, Functionality Testing, Security Testing, Session Management testing	Test Data would be required. "Basket" pages should be made and fully functional. Order page must be made and fully functional. The Models and Controllers must be created and there must be a logged in user to test it. The tester should place an order and fill out all the fields to check if valid data is accepted and invalid data gives error messages as well as make sure it successfully logs Product and displays it. They must be able to edit, delete and view the details of the Order. Test Data: Normal Products in basket: 2 "Strawberries" for £1.38, Delivery Method: Pickup, Address: 3 Caldmore Road.

			Email: Manual@gmail.com Pickup Date: 28/3/2026 Erroneous Product: Lenovo Laptop(Not a Product that exists in the system) Delivery Method: Nothing Address: 3 Nothing close Email: email(Must contain an @ symbol and must contain a domain e.g .com) Extreme Products in basket: - 3000 Pineapple for £4.60
20/4/26	Edit button in Order page	Blackbox Testing, Functionality integration	No test data required. The Orders Page must have already been created, and the Place Order functionality must have also been created. The account registration must have been created and there must be a logged in user to test it. The Tester should click details to ensure it loads up the correct page.
20/4/26	Delete button on the Order page	Blackbox testing, Functionality testing, integration testing	No test data required. Basket controller must be fully functional The Orders Page must have already been created, and the Place Order must have also been created. The account registration must have been created and there must be a logged in user to test it. The Tester should click details to ensure the user to delete the booking.
20/4/26	Details button on the Order Page	Blackbox testing, Functionality testing, integration testing	No test data required. The Order Page must have already been created, and the Place Order functionality must have also been created. Basket controller must be fully functional. The account registration must have been created and there must be a logged in user to test it. The Tester should click details to ensure the page loads up with all the correct booking details
20/4/26	Filter by Availability	Whitebox testing, black box testing ,	No test data required. The Orders page must

		Functionality Integration, Integration Testing	have already been created. The account registration must have been created and there must be a logged in user to test. The tester must click the filter by availability button and select date and the system must show all available Products to be Ordered on the specific date
20/4/26	“ADD” button on the products section of the homepage	Whitebox testing, black box testing , Functionality Integration, Integration Testing	No test data required. Home Page must be fully functional. The account registration must have been created and there must be a logged in user to test it. Migrations and update database must be applied. Basket controller must be made and fully operational. Tester must ne logged into a customer role account. Click Add button for any product in the product section to see if it will add to the basket with the right quantity.
20/4/26	Quantity change buttons on the products section of the home page	Whitebox testing, black box testing , Functionality Integration, Integration Testing	No test data required. Home Page must be fully functional. The account registration must have been created and there must be a logged in user to test it. Migrations and update database must be applied. Quantity change functionality must already be made. Tester must ne logged into a customer role account. The tester must then use + and – buttons to check if it increases or decreases the quantity. The tester must then add the changed quantity to basket and check if the correct amount of quantity for product has been added to the basket.
20/4/26	Product search Functionality	Whitebox testing, black box testing , Functionality Integration Testing	The products page must be created and fully functional and the search bar must be implemented. Products must be seeded in the database. The tester must type the search bar to verify if the products match the search and that no results returns an appropriate message. Test Data

			Normal: search: “Strawberries” Erroneous: Search: “\$\$\$£**”(special characters) Extreme: Search for 390 characters of random letters
20/4/26	Remove Item from Basket	Blackbox Testing, Functionality Testing, Integration Testing	The basket page and the basket controller must be fully operational. An registered or logged in user must add or have one item into their basket. The tester must interact with the remove button on an item in the basket and verify if it’s been removed, the basket automatically updates, and removing the last item in the basket triggers the system to display an empty basket message. Test Data Normal: Remove one of two items in the basket Erroneous: Search: Attempt to remove an item that is no longer in the database Extreme: Remove all items one by one and verify empty state.
20/4/26	Delivery Method selection at checkout	Blackbox Testing, Whitebox Testing, Functionality Testing	The checkout page must be created and fully functional. The Order model must be fully functional and created. The tester must verify that selecting a home address must require an address to be filled in and that the correct delivery charge is applied to the total amount of the order. Test Data Normal: Address, 3 Caldmore Road Erroneous: Address, Gum Time Extreme: 390 characters of random letters